$\boxed{\text{Example:}}$ $A = \begin{bmatrix} 1 & 2 & 4 \\ 0 & 0 & 5 \\ 0 & 3 & 6 \end{bmatrix}$ 3×3 Lecture 9

$a_1 \quad a_2 \quad a_3$

Can you find matrices Q & R using classical Gram Schmidt orthogonalization?

Sol^n:

QR factorization $\Rightarrow$ $\boxed{A = QR}$

$\rightarrow$ We want to find $\vec{q}_1, \vec{q}_2, \vec{q}_3$ using $\vec{a}_1, \vec{a}_2, \vec{a}_3$.

Let's find $\vec{q}_1, \vec{q}_2, \vec{q}_3$:

Step 1: Finding $\vec{q}_1$ using $\vec{a}_1$

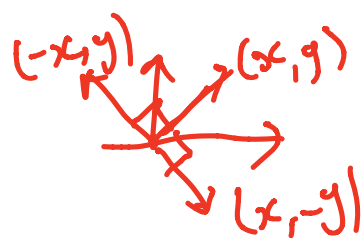
Let $\vec{v}_1 = \vec{a}_1 \Rightarrow \vec{v}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$.

Now find $\vec{q}_1 = \dfrac{\vec{v}_1}{\|\vec{v}_1\|} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$,

where $\|\vec{v}_1\| = \sqrt{1^2+0^2+0^2} = 1$

$\vec{q}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$

$v_1 = a_1 \Rightarrow q_1 = \dfrac{\vec{v}_1}{\|\vec{v}_1\|}$



Step 2: Finding $\vec{q}_2$ using $\vec{a}_2$ & $\vec{q}_1$

→ $\vec{q}_2$ should be orthogonal to $\vec{q}_1$

→ $\vec{q}_2$ should be of length "1"

$$\Rightarrow \vec{v}_2 = \vec{a}_2 - (\vec{a}_2^* \vec{q}_1)\vec{q}_1$$

start with $\vec{a}_2$

make $\vec{a}_2$ orthogonal to $\vec{q}_1$

(can also be viewed as orthogonal projection)

$$\begin{pmatrix} 2 & 0 & 3 \end{pmatrix}\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = 2$$

$$\Rightarrow \vec{v}_2 = \begin{bmatrix} 2 \\ 0 \\ 3 \end{bmatrix} - \boxed{2}\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 3 \end{bmatrix}$$

$\vec{a}_2^* \vec{q}_1$

$$\|\vec{v}_2\|_2 = \sqrt{0^2 + 0^2 + 3^2} = 3$$

$$\Rightarrow \vec{q}_2 = \frac{\vec{v}_2}{\|\vec{v}_2\|_2} = \frac{1}{3}\begin{bmatrix} 0 \\ 0 \\ 3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Step 3: Find $\vec{q}_3$ using $\vec{a}_3$ & $\vec{q}_1$ & $\vec{q}_2$

- $\vec{q}_3$ should be orthogonal to $\vec{q}_1$ & $\vec{q}_2$
- $\vec{q}_3$ should have length "1"

$$\vec{v}_3 = \underbrace{\vec{a}_3}_{\text{start with } \vec{a}_3} - \underbrace{(\vec{a}_3^* \vec{q}_1)\vec{q}_1}_{\text{make it orthogonal to } \vec{q}_1} - \underbrace{(\vec{a}_3^* \vec{q}_2)\vec{q}_2}_{\text{make it orthogonal to } \vec{q}_2}$$

Recall vector decomposition into orthogonal & non orthogonal components.

$$\vec{v}_3 = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix} - 4\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} - 6\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 5 \\ 0 \end{bmatrix}$$

$$\|\vec{v}_3\|_2 = \sqrt{0^2 + 5^2 + 0} = 5$$

$$\Rightarrow \vec{q}_3 = \frac{\vec{v}_3}{\|\vec{v}_3\|} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

Step 4: Combine $\vec{q}_1, \vec{q}_2, \vec{q}_3$ into Q as its matrix columns

$$\Rightarrow Q = \begin{array}{c} \begin{array}{ccc} q_1 & q_2 & q_3 \end{array} \\ \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \end{array}$$

$$A = QR \quad \Rightarrow \quad \underbrace{Q^* A} = R$$

Step 5: So what is $\underline{\underline{R}}$?

We know R is upper Δ

So let's backtrack a bit.

$$\boxed{r_{11}} \|v_1\| q_1 = \underbrace{\vec{a}_1}_{\vec{v}_1}$$

$$\boxed{r_{12}} \|v_2\| q_2 = \underbrace{\vec{a}_2 - (a_2^* \vec{q}_1)\vec{q}_1}_{\vec{v}_2}$$

$$\boxed{r_{33}} \|v_3\| q_3 = \underbrace{\vec{a}_3 - \underbrace{(a_3^* \vec{q}_1)}_{r_{13}}\vec{q}_1 - (a_3^* \vec{q}_2)\vec{q}_2}_{v_3}$$

r_{12}

r_{23}

the coefficients form the R matrix:

R, Q

$$\begin{bmatrix} r_{11} & r_{12} & r_{13} \\ & r_{22} & r_{23} \\ & & r_{33} \end{bmatrix} = R$$

Classical Gram-Schmidt Orthogonalization:

① Finds $\vec{q}_1, \vec{r}_{11}$

② Finds $\vec{q}_2, \vec{r}_{12}, \vec{r}_{22}$

③ Finds $\vec{q}_3, \vec{r}_{13}, \vec{r}_{23}, \vec{r}_{33}$

r_{nn}

numerically unstable!

Why?

Let's try & see this another way!

Modified Gram-Schmidt Orthogonalization

$a_1 = r_{11} q_1$

$a_2 = r_{12} q_1 + r_{22} q_2$

Uses orthogonality: $\underbrace{q_1^* q_2 = 0}$

$$\underbrace{q_1^* a_2} = r_{12} \underbrace{\vec{q}_1^* \vec{q}_1}_{1} + r_{22} \underbrace{\vec{q}_1^* \vec{q}_2}_{=0}$$

$$q_1^* a_2 = \underbrace{r_{22} q_1^* q}_{=} + \boxed{r_{12}}$$

$$r_{12} = q_1^* a_2$$

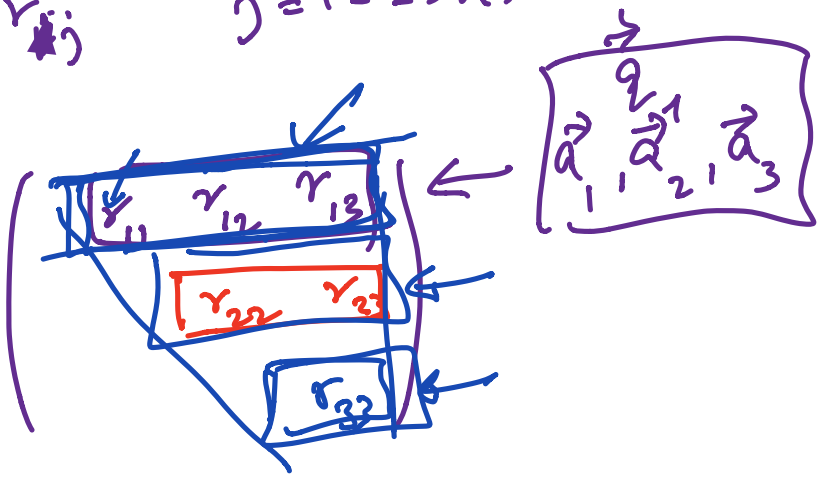
Next:

$$a_3 = r_{13} q_1 + r_{23} q_2 + r_{33} q_3$$

$$q_1^* a_3 = r_{13} \underbrace{q_1^* q_1}_{=1} + r_{23} \underbrace{q_1^* q_2}_{=0} + r_{33} \underbrace{q_1^* q_3}_{=0}$$

$$r_{13} = q_1^* a_3$$

We can use columns of $\vec{a}$ & $\vec{q}_1$ to r_{*j} $j = 1 \ldots n$.



2nd row:

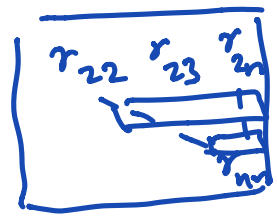
$$a_j = r_{1j} q_1 + \underbrace{\sum_{j=2}^{2} r_{ij} q_i}_{} \qquad 2 \le j \le n$$

Now let

$$\hat{a}_j = \vec{a}_j - r_{1j}\vec{q}_1 = \sum_{j=2}^{2} r_{ij} q_i$$

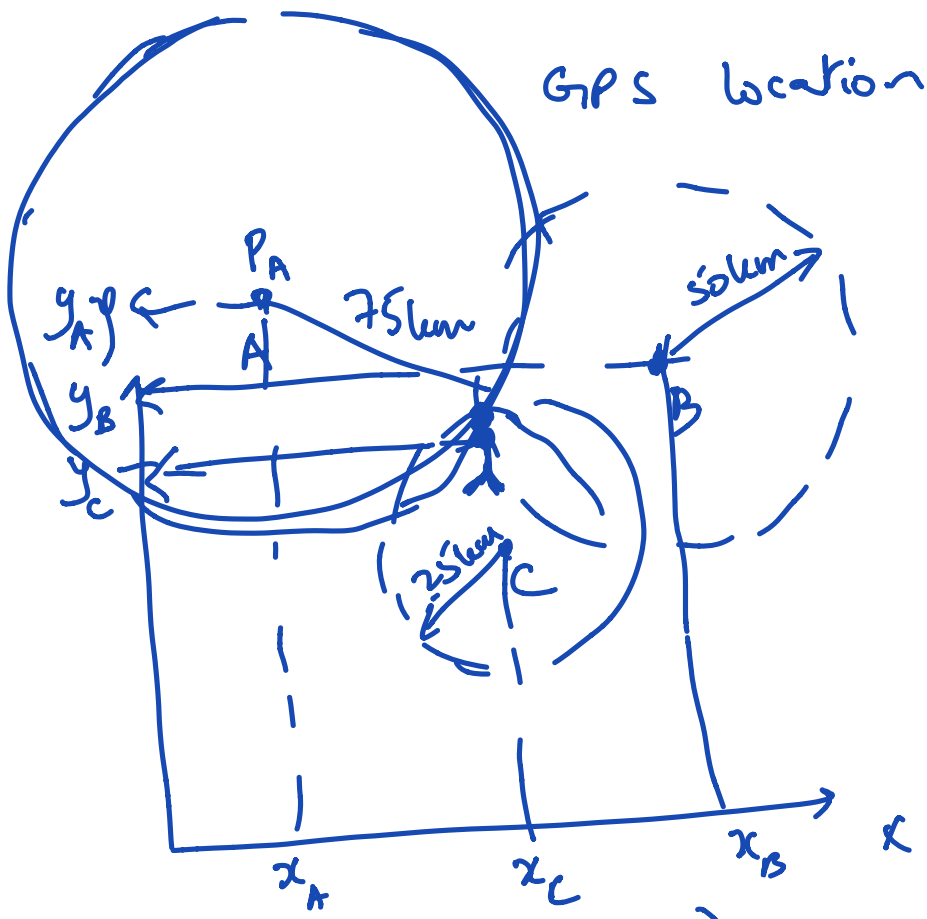
$$\begin{bmatrix} \hat{a}_2 & \hat{a}_3 & \hat{a}_n \end{bmatrix} = \begin{bmatrix} q_2 & q_3 & \cdots & q_n \end{bmatrix} \hat{R}$$

$(n-1)$



$(n-2)$

$n-3$



$$\begin{array}{c|ccc} & P_A & P_B & P_C \\ \hline x & x_A & x_B & x_C \\ y & y_A & y_B & y_C \end{array}$$

for S_A $\rightarrow (x - x_A)^2 + (y - y_A)^2 = 75^2$

for S_B $(\quad) + (\quad) = ?$

for S_C $(\quad) + (\quad) = ?$

$\Downarrow$

$$\underset{3\times 2}{\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ a_{31} & a_{32} \end{pmatrix}} \underset{2\times 1}{\begin{pmatrix} x \\ y \end{pmatrix}} = \underset{3\times 1}{\begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}}$$

$$\boxed{A \quad \vec{x} = \vec{b}}$$